

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\partial_\xi &= \frac{1}{d}\partial_q, \partial_\zeta = \frac{c\sqrt{\rho}}{d}\partial_p \\ \partial_\xi^2 &= \frac{1}{d^2}\partial_q^2, \partial_\xi\partial_\zeta = \frac{c\sqrt{\rho}}{d^2}\partial_p\partial_q, \partial_\zeta^2 = \frac{c^2}{d^2}\rho\partial_p^2 + \frac{1}{2}\frac{c^2}{d^2}\rho_p\partial_p,\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\mathcal{B}^a(x) = \begin{pmatrix} C^a(x) \\ D^a(x) \end{pmatrix}, \quad \mathcal{F}^a(x) = \begin{pmatrix} \chi^a(x) \\ \bar{\chi}^a(x) \\ \lambda^a(x) \\ \bar{\lambda}^a(x) \end{pmatrix},$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{bmatrix} g_2 + w_2^2 h_4 + n_2^2 h_5 & w_2 w_3 h_4 + n_2 n_3 h_5 & w_2 h_4 & n_2 h_5 \\ w_2 w_3 h_4 + n_2 n_3 h_5 & g_3 + w_3^2 h_4 + n_3^2 h_5 & w_3 h_4 & n_3 h_5 \\ w_2 h_4 & w_3 h_4 & h_4 & 0 \\ n_2 h_5 & n_3 h_5 & 0 & h_5 \end{bmatrix},$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}D^{++} &= u^{+\alpha} \frac{\partial}{\partial u^{-\alpha}}; & D^{--} &= u^{-\alpha} \frac{\partial}{\partial u^{+\alpha}} \\ 2D^{++} &= [D^0, D^{++}]; & -2D^{--} &= [D^0, D^{--}] \\ D^0 &= [D^{++}, D^{--}] = u^{+\alpha} \frac{\partial}{\partial u^{+\alpha}} - u^{-\alpha} \frac{\partial}{\partial u^{-\alpha}}\end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\zeta_\nu(s) = -\frac{1}{\pi} \Re \left\{ x^{1-s} e^{-i\frac{\pi}{2}s} \int_0^\infty (y-i)^{-s} \frac{\Delta_\nu^{TE'}(ix(y-i))}{\Delta_\nu^{TE}(ix(y-i))} dy \right\},$$